

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 24

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Modify the parallel Sieve of Eratosthenes program presented Lecture 24 to incorporate the first improvement: It should not set aside memory for even integers. Benchmark your program, comparing its performance with that of the original parallel sieve program.
2. Modify the parallel Sieve of Eratosthenes program presented in Lecture 24 to incorporate the first two improvements: Your program should not set aside memory for even integers, and each process should use the sequential Sieve of Eratosthenes algorithm on a separate array to find all primes between 3 and $\lfloor \sqrt{n} \rfloor$. With this information, the call to MPI_Bcast can be eliminated. Benchmark your program, comparing its performance with that of the original parallel sieve program.

- **Welcome: Sean**

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